

R22 CSE Subject Analysis

Purpose :

I reviewed the R22 B.Tech CSE curriculum to understand which subjects are still important for a student graduating around 2030 and which subjects need changes.

I have divided the subjects into three types:

- **KEEP** – The subject is still strongly relevant and its main concepts should remain.
- **UPDATE** – The basic concepts are important, but some topics, tools or practical parts need to be changed.
- **REDUCE** – The subject or some parts of it are less important and can be reduced.

For this analysis, I selected important core CSE subjects from different areas such as programming, systems, databases, algorithms, networking, DevOps, AI/ML and security.

1. CS305PC – Object Oriented Programming through Java

Semester: II Year – I Semester

Credits: 3

Decision: UPDATE

What is taught in R22?

- The subject covers:
- Classes and Objects
- Constructors
- Inheritance
- Polymorphism
- Method Overloading and Overriding
- Abstract Classes
- Interfaces
- Exception Handling
- Multithreading
- Generics
- AWT
- Swing
- Applets

My analysis :

Java and object-oriented programming are still useful. Concepts such as classes, objects, inheritance, interfaces, abstraction and polymorphism are important for understanding software development.

The problem is that some parts of the R22 syllabus are based on older Java technologies. Applets are no longer suitable for modern application development. AWT and Swing can still help understand GUI concepts, but they should not receive the same importance as modern application development practices.

What should change?

I would keep:

- OOP concepts
- Exception handling
- Collections and generics
- Multithreading
- Interfaces
- Modern Java features
- I would reduce or remove:
- Applets
- JApplet
- Old GUI-focused development

I would add more practical topics such as:

- Modern Java
- Maven or Gradle
- JUnit testing
- Git
- REST APIs
- JSON
- Debugging
- Building small real-world applications

My 2030 direction:

UPDATE – Modern Java Programming

The subject should teach Java as a modern programming language instead of focusing on old Java technologies.

2. CS302PC – Data Structures

Semester: II Year – I Semester

Credits: 3

Decision: KEEP

What is taught in R22?

The R22 syllabus covers important data structures such as:

- Arrays and Abstract Data Types
- Linked Lists
- Stacks and Queues
- Dictionaries and Hash Tables
- Binary Search Trees
- AVL Trees
- Red-Black Trees
- B/B+ Trees
- Graphs
- Sorting algorithms
- Pattern matching

- Tries

My analysis

I think Data Structures should definitely be kept because it is a basic subject for almost every CSE student. Data structures are not tied to one particular technology. They help students understand how data is stored, searched and processed efficiently.

The topics such as linked lists, stacks, queues, trees, graphs and hashing are still useful in software development, databases, operating systems and many other areas.

However, the subject can be made more practical. Instead of only learning the definitions and algorithms, students should implement them and understand where they are used in real applications.

What should change?

The core topics should remain, but more importance should be given to:

- Practical implementation
- Time and space complexity
- Real-world examples
- Problem-solving exercises
- Comparing different data structures
- Applications of graphs and hashing

My 2030 direction :

KEEP – Practical Data Structures

The concepts should remain almost the same, but teaching should become more application-oriented.

3. CS403PC – Operating Systems

Semester: II Year – II Semester

Credits: 3

Decision: UPDATE

What is taught in R22?

The R22 syllabus covers:

- Operating system structures
- System calls
- Processes
- Threads
- CPU scheduling
- Deadlocks
- Synchronization
- Inter-process communication
- Memory management
- Virtual memory
- Paging
- Segmentation

- File systems
- UNIX system calls

My analysis :

I think Operating Systems is a very important foundation subject and should not be removed.

Processes, threads, memory management, scheduling and file systems are fundamental concepts. Students need these concepts to understand how computers and software actually work.

However, students should also understand how these concepts are used in modern computing environments.

What should change?

More practical exposure should be given to:

- Linux
- Process and thread programming
- Virtual machines
- Containers
- Resource management
- Basic container isolation
- Modern system environments

The theory should remain, but students should be able to demonstrate the concepts practically.

My 2030 direction :

UPDATE – Modern Operating Systems and Systems

The foundation remains, but practical Linux, virtualization and container concepts should be included.

4. CS404PC – Database Management Systems

Semester: II Year – II Semester

Credits: 3

Decision: UPDATE

What is taught in R22?

R22 covers:

- Database concepts
- ER model
- Relational model
- Relational algebra
- SQL
- Views
- Triggers
- Functional dependencies
- Normalization
- Transactions
- Concurrency control
- Recovery
- Indexing

- B+ trees
- File organization

My analysis :

DBMS is still one of the most important subjects for CSE students. Almost every software application needs some form of data storage.

The relational model, SQL, normalization, transactions and indexing are still important. Therefore, I would not replace the core DBMS concepts.

The main improvement should be making the subject more connected to real applications.

What should change?

Students should also learn:

- Practical SQL
- Query optimization
- Execution plans
- Database APIs
- Cloud databases
- Basic NoSQL concepts
- Document databases
- Database security
- Working with databases from applications

My 2030 direction :

UPDATE – Modern Database Systems

The relational foundation should remain, while students should get exposure to modern database technologies and application development.

5. CS501PC – Design and Analysis of Algorithms

Semester: III Year – I Semester

Credits: 4

Decision: KEEP

What is taught in R22?

The subject focuses on:

- Algorithm analysis
- Complexity
- Searching
- Sorting
- Divide and Conquer
- Greedy methods
- Dynamic Programming
- Graph algorithms
- Optimization techniques

My analysis :

I would keep this subject because algorithmic thinking is a long-term CSE foundation.

Programming languages and tools may change, but the ability to analyse a problem and choose an efficient solution will continue to be important.

The main issue is not the syllabus itself. The teaching can be improved by connecting algorithms to real problems.

What should change?

More practical work should include:

- Coding problems
- Comparing algorithm performance
- Time and space analysis
- Benchmarking
- Real-world problem solving
- Choosing the right algorithm for a particular problem

My 2030 direction :**KEEP – Applied Algorithms**

The main concepts should remain, but practical problem-solving should receive more importance.

6. CS502PC – Computer Networks

Semester: III Year – I Semester

Credits: 3

Decision: UPDATE

What is taught in R22?

The syllabus covers:

- OSI model
- TCP/IP
- Physical and Data Link layers
- Error detection
- Sliding window protocols
- Ethernet
- Wireless LAN
- Routing
- Congestion control
- TCP
- UDP
- DNS
- HTTP
- Email
- Streaming

My analysis :

Networking fundamentals are still necessary. Students need to understand how computers communicate before they can properly understand cloud computing, distributed systems or network security.

Therefore, I would keep the basic networking concepts.

However, the syllabus should connect these concepts with the type of networking used in modern web and cloud applications.

What should change?

I would add more exposure to:

- IPv6
- Modern web protocols
- HTTP/2 and HTTP/3 concepts
- TLS
- Web APIs
- Cloud networking
- Virtual networks
- Network monitoring
- Basic software-defined networking concepts

My 2030 direction:

UPDATE – Modern Computer Networks

The fundamentals should remain, but students should understand networking in modern web and cloud environments.

7. CS503PC – DevOps

Semester: III Year – I Semester

Credits: 3

Decision: UPDATE

What is taught in R22?

The syllabus includes:

- Agile
- DevOps
- Continuous delivery
- Release management
- Scrum
- Kanban
- Git
- Docker
- Gerrit
- GitLab
- Jenkins
- Build pipelines
- Infrastructure as Code
- Selenium

- Ansible
- Chef
- Puppet
- Salt Stack

My analysis :

DevOps is highly relevant to software development because companies need to build, test and deploy software efficiently.

The R22 syllabus already includes several useful concepts such as Git, Docker and CI/CD. So I would not say that DevOps needs to be added from zero.

The main problem is that specific tools can become outdated. Therefore, students should learn the concepts and workflows rather than memorizing many tool names.

What should change?

More focus should be given to:

- Modern Git workflows
- CI/CD
- Containers
- Kubernetes concepts
- Cloud deployment
- Infrastructure as Code
- Monitoring and observability
- Automated testing
- DevSecOps

My 2030 direction :

UPDATE – Modern DevOps and Cloud-Native Development

The focus should move from learning many individual tools to understanding transferable DevOps practices.

8. CS601PC – Machine Learning

Semester: III Year – II Semester

Credits: 3

Decision: UPDATE

What is taught in R22?

The R22 syllabus includes:

- Supervised learning
- Unsupervised learning
- Linear regression
- Perceptron
- Neural networks
- Backpropagation
- SVM

- Decision trees
- Ensemble methods
- Boosting and bagging
- Clustering
- Dimensionality reduction
- Genetic algorithms
- Reinforcement learning
- Bayesian networks
- Hidden Markov Models

My analysis :

The mathematical and algorithmic foundations of Machine Learning are still important. Students should understand how models learn from data instead of only using ready-made AI tools.

However, modern ML work involves much more than training an algorithm. Students also need to know how data is prepared, how models are evaluated and how trained models are used in real applications.

What should change?

I would add:

- Python ML ecosystem
- Data preprocessing
- Feature engineering
- Model evaluation
- Cross-validation
- Experiment tracking
- ML pipelines
- Model deployment
- Monitoring
- Responsible AI concepts

My 2030 direction :

UPDATE – Machine Learning and ML Engineering

The existing ML algorithms should remain as the foundation, but the course should include the complete ML lifecycle.

9. CS603PC – Artificial Intelligence

Semester: III Year – II Semester

Credits: 3

Decision: UPDATE

What is taught in R22?

R22 already includes Artificial Intelligence as a core subject.

The subject provides the foundation for understanding intelligent systems and classical AI approaches.

My analysis

I would not say that AI is completely missing from the R22 curriculum because AI is already present as a core subject.

The bigger issue is that the AI curriculum needs to reflect how AI systems are being developed today.

Students should first understand classical AI concepts and then connect them to modern AI approaches.

What should change?

The course should include concepts such as:

- Neural-network-based AI
- Deep learning
- Embeddings
- Transformers
- Generative AI
- Large language models
- Prompt engineering
- AI evaluation
- Responsible AI
- AI application development

The important point is that these topics should be added by updating the existing AI course rather than simply creating another general "AI" subject.

My 2030 direction:

UPDATE – Modern AI

The R25 curriculum already strengthens AI-related education, so my proposal should explain what is still missing rather than simply saying "add AI."

10. CS701PC – Cryptography and Network Security

Semester: IV Year – I Semester

Credits: 3

Decision: UPDATE

What is taught in R22?

The syllabus includes:

- Security concepts
- Security attacks
- Security services
- Substitution and transposition techniques
- Symmetric cryptography
- Asymmetric cryptography
- DES
- AES
- Blowfish
- RC5
- RSA
- Diffie-Hellman

- Hash functions
- HMAC
- Digital signatures
- Key management
- Kerberos
- X.509
- HTTPS
- SSH
- IPsec
- Email security
- Web security

My analysis :

I think the security foundation is still very important. Students should understand encryption, hashing, authentication, digital signatures and key management.

At the same time, modern software security involves more than traditional cryptographic algorithms.

Students increasingly work with web applications, APIs, cloud services and distributed systems, so security should be connected to these environments.

What should change?

I would add more practical topics such as:

- TLS concepts
- Secure API design
- Identity and Access Management
- Cloud security
- Application security
- Authentication
- Threat modelling
- Vulnerability management
- Secure software development
- DevSecOps

Older algorithms can still be taught for understanding and comparison, but they should not take most of the practical focus.

My 2030 direction :

UPDATE – Modern Cryptography, Application and Cloud Security

The existing security foundation should remain, while application, cloud and software-security practices should receive more importance.

Overall Subject Analysis

Course	Subject	Decision
302PC	Data Structures	KEEP

ourse	Subject	Decision
305PC	OOP through Java	UPDATE
403PC	Operating Systems	UPDATE
404PC	DBMS	UPDATE
501PC	Design and Analysis of Algorithms	KEEP
502PC	Computer Networks	UPDATE
503PC	DevOps	UPDATE
601PC	Machine Learning	UPDATE
603PC	Artificial Intelligence	UPDATE
701PC	Cryptography and Network Security	UPDATE

Main observation :

My main conclusion is that the R22 curriculum does not need to be completely replaced. Many foundation subjects are still valuable.

The bigger requirement is to **update the parts that depend on old technologies and add modern practical skills around the existing foundations.**

This approach avoids removing important CSE fundamentals while also preparing students for current software development and future technologies.